

VPN-LAN Service Access — Status Summary

Revision: 3 **Last modified:** 2026-07-01T19:05:00Z **Status:**
Companion summary of [Status.md](#) (§11.4.56 two-audience).

Page 1 — For the operator / stakeholders (plain language)

We are building `helix_proxy`'s ability to **reach and use the services on the far side of the VPN** — file shares, file transfer, email, casting, and Android devices. The design is finished and security-checked, and the plumbing that connects to the VPN (via your existing `sword_toolkit` scripts) is built and proven.

What's done and proven right now (without needing the live VPN):

- **The bridge connector** — `helix_proxy` reads where your `sword_toolkit` scripts live from a private settings file, checks whether the VPN is up with a "doctor" tool, and — crucially — when the VPN is **down** it **honestly reports "skipped," never a fake success**. The test suite confirms this and also proves the check has real teeth (a healthy stub flips it to "up"). This is the guarantee that we never pretend a feature works when it doesn't.
- **The protocol tests** — the file-share (SMB/NFS), file-transfer (FTP/SFTP/WebDAV), and email tests are written and self-proven: each writes and reads back real data (checked byte-for-byte) when the VPN is up, and the email test includes a **spam-relay safety check** that would fail `helix_proxy` if it ever accepted unauthenticated mail. When the VPN is down, they all skip honestly.
- **Miracast** is honestly marked **impossible** over this kind of VPN (it's a direct radio link, not an internet protocol) — with **Chromecast** offered as the working alternative.
- **The operator guide** is written (setup steps, protocol table, FAQ).

What needs you (clearly flagged, we will ask before acting):
the **live VPN connection** (your secrets + Mullvad + admin rights) unlocks the real end-to-end tests; deploying the

device-**discovery helper** on a remote machine, and **flashing** any real device, each need your go-ahead first — we never change sword_toolkit or a remote host without asking.

Bottom line: the feature is designed, the anti-bluff plumbing is built and proven, and the live parts switch on the moment you connect the VPN. Nothing is faked; skipped means skipped.

Page 2 — For software engineers

L3-routed VPN ⇒ route unicast / proxy HTTP-shaped via Squid / reflect multicast / L2 structurally-impossible. Env-var bridge (§11.4.28), honest-SKIP gate (bridge_require rc 2 OPERATOR-BLOCKED §11.4.68), anti-bluff ab_pass_with_evidence (non-empty artefact required).

Phase	Scope	Status	Commit / evidence
0	bridge scaffold + sword-doctor	PASS	d781002 + suite test_vpn_lan_bridge GREEN (§11.4.115 teeth) 5c28f56
1	routed gateway + SSRF reconciliation	AUTHORED (local-stub GREEN)	1d87b3a ssrf_carveout_teeth.sh: floor survives narrow carve (T1/T2/T3), SSRF_MUT=1 catches 2/2 collapses; wired into standing suite; live carve bridge+operator-gated (§11.4.120/§11.4.133)
2	SMB/CIFS/ NMB + NFS	AUTHORED (SKIP-proven)	182e80a sha256 round-trip; bridge-down PASS_lines=0
3	FTP/FTPS/ SFTP + WebDAV(via Squid)	AUTHORED (SKIP-proven)	182e80a wrong-answer⇒FAIL
4	IMAP/SMTP-submission/ POP3 + open-relay guard	AUTHORED (SKIP-proven)	2f31460 external-RCPT-accepted⇒FAIL
5	mDNS/SSDP/ WS-Disc/ DNS-SD reflector	AUTHORED (SKIP-proven)	d0b42df reflector_design.md (cited RFCs) + discovery_reflect.sh; remote deploy operator-gated (§11.4.122)
6	Chromecast/ DIAL	AUTHORED (SKIP-proven)	65043ce chromecast_dial.sh; eureka 8008 + CASTV2 8009

Phase	Scope	Status	Commit / evidence
			transition (§11.4.107); discovery = Phase-5 dep
7	ADB (access/debug/connect/flash)	AUTHORED (SKIP-proven)	65043ce adb_over_vpn.sh; routed 5555 + §11.4.174 serial-safety; usbip flash operator-gated (§11.4.133)
8	Miracast	PASS (Won't-fix, §11.4.112)	12faf12 cited verdict + Cast alternative
9	operator bridge-setup guide	PASS	a5e5616 leak-clean HTML+PDF
10	containerize (§11.4.76)	AUTHORED (SKIP-proven)	1911d5e containerization.md + vpn_lan_containers.yaml + container_boot.sh; submodule untouched
11	§11.4.169 coverage + Challenges + HelixQA	AUTHORED (mostly GREEN)	89f73b7 Challenge + 9-case bank; autonomous types GREEN — stress+chaos 97d9733, bench+memory + dns-rebinding 4e2810c, concurrency+load 7e73b08; HelixQA-run blocked 6 siblings+podman (§11.4.3)
12	Bidirectional exposure + ingress allowlist	AUTHORED (GREEN + suite-wired)	2ed0fed bidirectional_exposure.md + ingress_allowlist_teeth.sh (INGRESS_MUT 3/3, PASS=4); cc8b620 both-way reverse-leg in all 5 protocol tests; live both-way operator-gated
—	§11.4.135 autonomous suite guards	GREEN	6d8893f test_vpn_lan_ssrf + test_vpn_lan_ingress + test_vpn_lan_autonomous_battery wired, all rc=0 PASS≥1

Autonomous value delivered: the anti-bluff gate (bridge-down ⇒ honest SKIP, exit 0, zero ^PASS:) + the §11.4.115 teeth in the standing suite. Live round-trip evidence (bytes/sha256/mailbox/serials) is operator-gated on the sword connection. Composes §11.4.3/§11.4.28/§11.4.45/§11.4.68/§11.4.69/§11.4.108/§11.4.112/§11.4.115/§11.4.120/ §11.4.122/§11.4.133/§11.4.167/§11.4.169.